Mutation Tester Usability Test Analysis

Results

Task	Participant 1	Participant 2	Participant 3
1. Install and run	5 min	6 min	30 min, stuck at step 3
2. Read the report	Wrong counts; couldn't define the terms	4 min; unsure which test caught each mutant	Couldn't find NoCoverageDemo.py
3. Cover untested code	Couldn't find NoCoverageDemo.py	8 min	6 min
4. Investigate a survivor	Edited the config, couldn't run it	6 min	7 min
5. Choose operators	Not reached	"I didn't know what I was doing," skipped ahead	Skipped
6. Add a rule to the spec	Not reached	5 min	4 min
7. Check a risky mutant	Not reached	No time given	Not reached

Findings

1. **Two of the three opened the wrong report.**
    a. Setup runs the safety demo right before it says to open the report, so Participants 1 and 3 most likely opened the safety demo's. That explains the wrong counts and the missing NoCoverageDemo.py, and it blocked Tasks 2 and 3 for both.
2. **Nothing tells you the operator names.**
    a. The doc never lists names like AOR_PLUS_TO_MINUS. Participant 2 called Task 5 the hardest and skipped it.
3. **Running with a different config isn't explained.**
    a. Windows named the copy newPython - Copy.config, and the spaces break the command unless it's quoted.
4. **The concepts didn't land for everyone.**
    a. Participant 1 couldn't define killed or survived: "I still have no idea what this program is for." Participant 2 couldn't tell which test caught each mutant.
5. **Safety prompts get repetitive.**
    a. Participant 2 asked for a "Continue all" option.

What worked: Participants 2 and 3 found it clear which line each mutant changed ("a strength of the program"). Participant 1 only found it by hovering.

Next steps

- **Instructions:** move the safety demo into Task 7, list the operator names, show the command for running a different config, add a `cd` before the clone, and put the intro before setup.
- **Tool:** a clear error when the config isn't found, a "Continue all" choice at safety prompts, a legend for the three states, and a hover hint in the report.
- **Test format:** read-only instructions plus a separate answer form that requires an answer per task (one participant typed over the build command; another left every answer blank).